Paper Code: MGI 105

M.Tech. Geo-Informatics

End Semester Examination-2024

1st Semester

DIGITAL IMAGE PROCESSING

Time: 03:00 hrs **MM: 100**

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question is twenty.
(iv) Each question carries 10 marks.

Q1.		**(20 Marks)**
(a)	A 16 bit digital image contains 500 rows and 1000 columns. Determine the space required to store this digital image in Kb and Mb. If instead of using 16 bit image, 10 bit image is used then determine the space required to store this digital image in Kb and Mb.	**CO1**
(b)	What do you understand by the detector error? List the different type of detector error and the method to remove them.	**CO1**
(c)	What do you understand by digital images? How these images are formed? What are the different univariate statistics used in remote sensing image analysis?	**CO1**
Q2.		**(20 Marks)**
(a)	Explain scattering and absorption in brief and its significance in optical, IR and thermal IR band.	**CO2**
(b)	In radiometric correction explain the requirement of correcting the slope and aspect effects. Describe at least two methods used for correcting the slope effect by showing proper diagram.	**CO2**
(c)	What are the important spectrum band that are used in remote sensing? How they are important in analysing the remote sensing images. Provide a brief discussion on NDVI.	**CO2**
Q3.		**(20 Marks)**
(a)	What is the requirement of contrast enhancement? Jusify your answer with some example.	**CO3**
(b)	Draw the histogram for the following image.	**CO3**

13	14	5	6	7	3	4
15	12	7	9	8	5	2
12	12	7	10	10	2	2
14	13	8	9	9	4	3
11	15	7	8	11	1	5
14	11	6	11	8	4	1

What type of information we can draw from histogram and how it is useful in remote sensing image analysis.

(c)	Following is the 4-bit 4×4 image and a 3×3 filter mask to be applied. Determine the result image without applying the padding and after applying the padding.	**CO3**

Image I:

7	9	15	15
7	10	14	12
8	9	12	14
7	8	11	11

Filter Mask

1	2	1
2	3	2
1	2	1

Q4. **(20 Marks)**

(a) Differentiate between supervised and unsupervised classification, parametric and non-parametric classification and soft and hard classifier. **CO4**

(b) Explain the minimum distance to mean classification by considering the Euclidean Distance. By taking an example of three band and three land cover classes of your choice show the assignment of a pixel to the land cover class. **CO4**

(c) Explain parallelepiped classification algorithm by taking the example of two band images and considering the pixel values of your choice. **CO4**

Q5. **(20 Marks)**

(a) Explain the term: Radiant flux, reflectance, absorptance, transmittance, irradiance, exitance, and radiance. **CO5**

(b) With respect to accuracy assessment explain the following terms (i) Ground Data Collection (ii) Spatial Autocorrelation (iii) Sample Size, (iv) Sampling Scheme. **CO5**

(c) Determine the overall accuracy, user and producer accuracy for the following confusion matrix. **CO5**

	Urban	Lake	Agriculture	Forest
Urban	30	1	4	8
Lake	2	40	6	5
Agriculture	6	6	30	3
Forest	4	5	3	50